

Task Manager with User Authentication — Submission

Writeup:

Project: Task Manager with User Authentication. CLI app with registration, login, task create/view/complete/delete. Data persisted in task_manager_data.json; passwords hashed with SHA-256.

Screenshots to attach:

- login_screen.png
- add_task.png
- task_list.png

Source Code:

```
import json
import os
import hashlib

class TaskManager:
    def __init__(self, data_file="task_manager_data.json"):
        self.data_file = data_file
        self.users = {} # {username: hashed_password}
        self.tasks = {} # {username: [task_dicts]}
        self.current_user = None
        self.load_data()

    # --- FILE HANDLING ---
    def load_data(self):
        if os.path.exists(self.data_file):
            try:
                with open(self.data_file, "r", encoding="utf-8") as file:
                    data = json.load(file)
                    self.users = data.get("users", {})
                    self.tasks = data.get("tasks", {})
            except json.JSONDecodeError:
                print("■■ Storage file corrupted. Starting with an empty dataset.")
                self.users = {}
                self.tasks = {}

    def save_data(self):
        data_to_save = {"users": self.users, "tasks": self.tasks}
        with open(self.data_file, "w", encoding="utf-8") as file:
            json.dump(data_to_save, file, indent=4, ensure_ascii=False)

    def _hash_password(self, password):
        return hashlib.sha256(password.encode("utf-8")).hexdigest()

    # --- AUTH ---
    def register(self):
        print("\n--- User Registration ---")
        username = input("Enter a new username: ").strip()
        if not username:
            print("■ Username cannot be empty.")
            return
        if username in self.users:
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        print("■ Username already exists.")
        return
password = input("Enter a password: ").strip()
if not password:
    print("■ Password cannot be empty.")
    return
self.users[username] = self._hash_password(password)
self.tasks.setdefault(username, [])
self.save_data()
print(f"■ User '{username}' registered successfully!")

def login(self):
    print("\n--- User Login ---")
    username = input("Username: ").strip()
    password = input("Password: ").strip()
    if username in self.users and self.users[username] == self._hash_password(password):
        self.current_user = username
        print(f"■ Login successful! Welcome, {username}.")
        return True
    print("■ Invalid username or password.")
    return False

def logout(self):
    if self.current_user:
        print(f"\nLogging out '{self.current_user}'...")
        self.current_user = None
    else:
        print("No user currently logged in.")

# --- TASKS ---
def add_task(self):
    if not self.current_user:
        print("■ You must be logged in to add tasks.")
        return
    print("\n--- Add New Task ---")
    description = input("Enter task description: ").strip()
    if not description:
        print("■ Task description cannot be empty.")
        return
    user_tasks = self.tasks.get(self.current_user, [])
    next_id = (max((t["id"] for t in user_tasks), default=0) + 1)
    new_task = {"id": next_id, "description": description, "status": "Pending"}
    user_tasks.append(new_task)
    self.tasks[self.current_user] = user_tasks
    self.save_data()
    print(f"■ Task #{next_id} added.")

def view_tasks(self):
    if not self.current_user:
        print("■ You must be logged in to view tasks.")
        return
    print("\n--- Your Tasks ---")
    user_tasks = self.tasks.get(self.current_user, [])
    if not user_tasks:
        print("■ No tasks found.")
        return
    print(f"{'ID':<6}{'Description':<40}{'Status':<12}")

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print("-" * 60)
for t in user_tasks:
    print(f"{t['id']:<6}{t['description'][:38]:<40}{t['status']:<12}")

def mark_task_completed(self):
    if not self.current_user:
        print("■ You must be logged in to modify tasks.")
        return
    try:
        tid = int(input("Enter task ID to mark completed: ").strip())
    except ValueError:
        print("■ Invalid ID.")
        return
    user_tasks = self.tasks.get(self.current_user, [])
    for t in user_tasks:
        if t["id"] == tid:
            if t["status"].lower() == "completed":
                print("■■ Task already completed.")
            else:
                t["status"] = "Completed"
                self.save_data()
                print(f"■ Task #{tid} marked as completed.")
            return
    print("■ Task ID not found.")

def delete_task(self):
    if not self.current_user:
        print("■ You must be logged in to delete tasks.")
        return
    try:
        tid = int(input("Enter task ID to delete: ").strip())
    except ValueError:
        print("■ Invalid ID.")
        return
    user_tasks = self.tasks.get(self.current_user, [])
    for i, t in enumerate(user_tasks):
        if t["id"] == tid:
            confirm = input(f"Type 'YES' to delete task #{tid}: ").strip()
            if confirm == "YES":
                user_tasks.pop(i)
                self.tasks[self.current_user] = user_tasks
                self.save_data()
                print(f"■ Task #{tid} deleted.")
            else:
                print("Canceled.")
            return
    print("■ Task ID not found.")

def main():
    mgr = TaskManager()
    while True:
        if mgr.current_user:
            print("\n--- Task Manager (logged in as: {}) ---".format(mgr.current_user))
            print("1. Add Task")
            print("2. View Tasks")
            print("3. Mark Task Completed")
            print("4. Delete Task")
            print("5. Logout")

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    print("6. Exit")
    choice = input("Choose (1-6): ").strip()
    if choice == "1":
        mgr.add_task()
    elif choice == "2":
        mgr.view_tasks()
    elif choice == "3":
        mgr.mark_task_completed()
    elif choice == "4":
        mgr.delete_task()
    elif choice == "5":
        mgr.logout()
    elif choice == "6":
        print("Exiting... Goodbye!")
        break
    else:
        print("■ Invalid choice.")
else:
    print("\n--- Task Manager ---")
    print("1. Register")
    print("2. Login")
    print("3. Exit")
    choice = input("Choose (1-3): ").strip()
    if choice == "1":
        mgr.register()
    elif choice == "2":
        mgr.login()
    elif choice == "3":
        print("Exiting... Goodbye!")
        break
    else:
        print("■ Invalid choice.")

if __name__ == "__main__":
    main()

```

Additional Remarks:

This CLI stores hashed passwords and task data locally in task_manager_data.json (UTF-8).

Register to create a user, then Login to manage tasks. Files to attach:

TaskManagerUserAuthentication.py, screenshots.